The Formulation Design Space for Execution-Free Code Verification: A Systematic Empirical Study

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Abstract

LLM-based program repair increasingly relies on generating multiple candidate patches and selecting the best via a verifier. Execution-based verification is reliable but expensive; three execution-free alternatives have emerged: per-test binary classification (CLS), execution output generation (CWM), and trajectory-level patch scoring (TRAJ). No prior work compares these formulations under matched conditions.

We present the first systematic comparison across two representative code LLM families (Qwen3, DeepSeek-Coder), two complexity levels, and multiple seeds under $\leq$8B LoRA fine-tuning. At function level, CLS and CWM are statistically equivalent across both families (TOST, 5 seeds): classification is not inherently easier than generation when outputs are short. TRAJ parity is model-dependent. At repository level, a significant interaction ($z = -23.7$) confirms this equivalence collapses: the CLS–CWM gap widens to 19–26 pp in single-seed variant-level evaluation, driven by format collapse in generation-based simulation. Gap magnitude depends on multiple confounds.

Error complementarity and a failure mode taxonomy motivate complexity-aware formulation routing over uniform selection.

1 Introduction

LLM-based automated program repair systems increasingly rely on test-time compute scaling: generating K candidate patches and selecting the best via a verifier (Snell et al., 2025). Full execution verification is reliable but expensive ($1–3 per candidate at repository scale[1]), driving demand for execution-free alternatives. Three distinct formulations have emerged for this task: **generation-based simulation** (FAIR CodeGen team et al., 2025; Maimon et al., 2026) predicts execution output strings per test; **trajectory-level scoring** (Shum et al., 2025) predicts a binary patch verdict without per-test decomposition; and **per-test classification** (Ni et al., 2023; Chen et al., 2023) predicts a pass/fail token per test. Each makes different tradeoffs between diagnostic granularity, output complexity, and inference cost.

Despite active development along each axis, no prior work provides a systematic comparison holding model capacity, training data, and compute constant. Practitioners choosing a formulation for their verification pipeline must rely on results from different models, different datasets, and different evaluation protocols. Whether one formulation is uniformly preferable, or whether the choice depends on properties of the code being verified, remains an open empirical question. A related question is whether formulation choice matters at all relative to more basic decisions: at what complexity level does fine-tuning become necessary, and does formulation selection add value beyond what fine-tuning alone provides?

We present the first systematic empirical comparison across two representative code LLM families (Qwen3, DeepSeek-Coder), two complexity levels (function-level and repository-level), and multiple seeds. Our contribution is primarily methodological: a matched experimental framework that isolates formulation effects from confounding variables. The analysis also produces actionable outputs: a failure mode taxonomy that characterizes when and why each formulation fails (§3.5), error complementarity findings with practical ensemble implications (§3.6), and formulation selection recommendations conditioned on code complexity (§5). The results reveal a three-layer structure. Fine-tuning is necessary (0-shot→fine-tuned gains of 35–36 pp at repo level). After fine-tuning, *formulation choice is governed by code complexity*: at function level, all formulations

[1]Based on per-instance costs reported by Yang et al. (2024) and Xia et al. (2025).

achieve comparable accuracy—consistent with the theoretical analysis in §2.2, which predicts that short outputs minimize generation-specific failures. At repository level under our ≤8B LoRA protocol, this equivalence collapses; variant-level analysis reveals three distinct failure modes consistent across model families, though gap magnitude depends on multiple confounds (§3.7). Figure 1 illustrates the three formulations.

We make the following contributions:

- **Formulation near-equivalence at low complexity.** At function level, CLS and CWM achieve statistical equivalence across two representative code LLM families; TRAJ reaches parity on Qwen3 but exceeds the equivalence margin on DeepSeek-Coder. Per-test classification matches trajectory-level accuracy while identifying which tests fail (§3.2, §3.4).
- **Complexity-dependent divergence.** A significant interaction ($z{=}-23.7$, $n{=}56{,}480$) confirms that formulations equivalent at function level (≤1.3 pp) diverge by 19–26 pp at repository level under ≤8B LoRA fine-tuning, driven by format collapse—four targeted interventions all fail to close the gap (Appendix H(g)). Variant-level analysis reveals three failure modes consistent across model families, though gap magnitude depends on single-seed evaluation and multiple confounds (§3.5, §3.7).
- **Error complementarity.** CLS and CWM exhibit partially complementary error patterns at function level (oracle: 90.63%; single-seed); at repo level, oracle routing reaches 95.83% variant accuracy (+14.58 pp; single-seed), the strongest practical argument for exploring the formulation design space (§3.6).

2 Method

Let p denote a candidate patch and $T = \{t_1, \ldots, t_n\}$ its test suite, where each t_i has label $y_i \in \{\text{PASS}, \text{FAIL}\}$ from execution. Tests decompose into *fail-to-pass* (T_{f2p}) and *pass-to-pass* (T_{p2p}) subsets; a patch is correct iff $\forall\, t_i$: $y_i = \text{PASS}$. We compare three formulations for predicting correctness without execution.

2.1 Verification Formulations

Let c_p denote the code context (unified diff) of patch p.

CLS (per-test classification). $f(c_p, t_i) \to \hat{y}_i \in \{\text{PASS}, \text{FAIL}\}$ predicts a single binary token per test (cross-entropy loss). Patch verdict: $\hat{y}_p = \bigwedge_i [\hat{y}_i = \text{PASS}]$. Output cost: one token per test.

CWM (output generation). $g(c_p, t_i) \to \hat{o}_i \in \Sigma^*$ generates the execution output string (language modeling loss). Verdict by exact match: $\hat{y}_i = \text{PASS}$ iff $\hat{o}_i = o_i^*$.[2] When the generated output is unparseable (i.e., does not conform to the expected structured format), the prediction defaults to the majority class verdict. Output cost: $\sum_i |\hat{o}_i|$ tokens.

TRAJ (trajectory-level). $h(c_p, T) \to \hat{y}_p \in \{\text{RESOLVED}, \text{UNRESOLVED}\}$ predicts a single patch verdict from all tests and provides no per-test signal. TRAJ serves as a granularity control: CLS–TRAJ differences isolate the value of per-test decomposition independent of the classification formulation itself.

2.2 Formulation-Specific Failure Modes

Each formulation has characteristic failure modes that differ in their sensitivity to code complexity:

Generation failure modes (CWM). The open-ended output space introduces three failure modes: (1) *token-budget exhaustion*, where $|o_i^*| > B$ guarantees exact-match failure due to truncation; (2) *string-value prediction errors*, where semantically equivalent outputs differing in surface form (floating-point precision, dictionary ordering) produce false negatives; and (3) *format compliance collapse*, where the model fails to produce structured predictions entirely at high input complexity, even when the generation budget is sufficient. Under optimistic independence, a token sequence of length L with per-token accuracy q has correct-sequence probability q^L; long outputs are therefore exponentially harder to predict exactly. We adopt exact string matching following the original CWM evaluation protocol (FAIR CodeGen team et al., 2025). Unparseable predictions default to the majority-class label (PASS).

Classification failure modes (CLS). The single-token output space eliminates format and truncation failures but constrains the model to encode its reasoning entirely within internal

[2] Exact string matching is conservative, consistent with FAIR CodeGen team et al. (2025). Semantic equivalences are not counted.

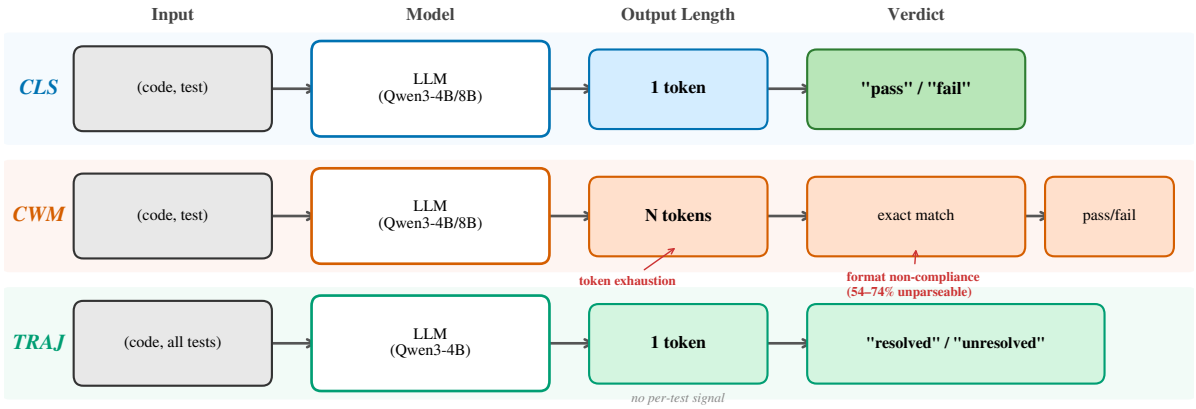


Figure 1: Three verification formulations. **Left:** CLS predicts binary pass/fail per test (1 output token). **Center:** CWM generates execution output, derives verdict by exact match ($O(L)$ tokens). **Right:** Trajectory-level scoring predicts patch-level resolved/unresolved (1 token, no per-test signal). Each formulation makes different tradeoffs between output complexity, diagnostic granularity, and inference cost.

representations, without the intermediate reasoning trace that generation provides. CLS errors manifest as false positives or false negatives without format failures; at function level, a mild FP bias (56.5%) emerges, with the error direction depending on class distribution and fine-tuning dynamics.

Trajectory-level failure modes (TRAJ). By collapsing all per-test information into a single binary verdict, TRAJ discards diagnostic granularity. When a patch fails multiple tests, the trajectory-level formulation cannot identify *which* tests fail. This limits downstream utility (e.g., targeted re-repair) but avoids per-test aggregation errors.

Complexity dependence. These failure modes suggest a complexity-dependent interaction: when execution outputs are short and structured (function-level code), generation modes (1)–(2) are rarely triggered, and all formulations should perform comparably. As output length and structural complexity increase (repository-level code), generation-specific failures may compound, creating divergence. Whether this divergence reflects a fundamental property of the generation formulation or is contingent on model/tokenizer properties is an empirical question we investigate in §3.

2.3 Oracle Accuracy-Utility Framework

For a buggy patch with $k = |T_{\text{f2p}}|$ fail-to-pass tests, a per-test verifier with recall r detects the bug if it identifies at least one failing test: $P_{\text{detect}}(k) = 1 - (1 - r)^{k_{\text{eff}}}$, where $k_{\text{eff}} = k/(1 + (k - 1)\rho)$ accounts for within-patch correlation via ICC ρ (Shrout and Fleiss, 1979). The per-test advantage over a trajectory-level baseline (a_h) is $\Delta(k) = P_{\text{detect}}(k) - a_h$, with population expectation $\mathbb{E}[\Delta] = \sum_k P(|T_{\text{f2p}}| = k) \cdot \Delta(k)$. This predicts that per-test advantage concentrates in multi-F2P-test instances and vanishes at $k{=}1$, validated in §3 and Appendix A.

3 Experiments

We systematically compare CLS, CWM, and TRAJ across two representative code LLM families, two complexity levels, and multiple seeds to characterize the formulation design space for neural code verification.

3.1 Experimental Setup

Datasets. *Function-level*: mutated variants from HumanEval (Chen et al., 2021) and MBPP (Austin et al., 2021) (five mutation types), labels verified by execution. 19,076 samples (train 15,290 / val 1,896 / test 1,890; pass:fail ≈ 52:48, majority baseline 51.75%). *Repo-level*: per-test instances from SWE-bench Verified (Chowdhury et al., 2024) mutants. ∼70K training samples; val ($n{=}9{,}335$) and test ($n{=}9{,}784$) have zero problem overlap (majority baseline 72.96%). Val-based checkpoint selection follows standard practice (Jimenez et al., 2024) where validation loss is computable; runs using truncate preprocessing produce NaN validation loss due to length-mismatched sequences, which defaults to the final epoch checkpoint (affects DeepSeek CLS s123,

all CWM runs, and Qwen3 TRAJ runs; see Appendix E).

Models and training. We fine-tune Qwen3-4B and Qwen3-8B (Yang et al., 2025a) with Low-Rank Adaptation (LoRA) (Hu et al., 2022) (rank 16, 3 epochs) across five seeds. Seeds 42/123/456 use bfloat16; seeds 789/0 use NormalFloat 4-bit (NF4) quantization (Dettmers et al., 2023) to test robustness to precision reduction (function-level accuracy remains within seed noise across both precisions; see footnote in §3.4). As a cross-model consistency check, we additionally fine-tune DeepSeek-Coder-6.7B (Guo et al., 2024) with identical LoRA configuration (2 seeds each for CLS and CWM). Our CWM baseline is a LoRA fine-tuned variant for matched comparison, not a reproduction of the 32B Code World Model (FAIR CodeGen team et al., 2025). Repo-level models use val-selected checkpoints where available; runs with NaN validation loss use the final epoch checkpoint (Appendix E). Zero-shot baselines use Qwen3-8B and Qwen3-32B (NF4) as LLM-as-judge. Full hyperparameter details appear in Appendix E.

Statistics. Confidence intervals use clustered bootstrap (Davison and Hinkley, 1997) (114 problem clusters, 10K resamples; we resample entire problems, not individual tests, to preserve within-problem correlations); per-test vs. trajectory-level equivalence uses Two One-Sided Tests (TOST) (Schuirmann, 1987) with ±2 percentage point (pp) margin. Holm-Bonferroni correction (Holm, 1979) controls family-wise error rate across the four primary comparisons (Table 15). Intraclass correlation coefficient (ICC) (Shrout and Fleiss, 1979) quantifies within-problem correlation. Oracle analysis uses SWE-bench Verified (500 instances).

3.2 Function-Level Results

Table 1 reports function-level results across formulations and model families.

All three formulations achieve comparable accuracy on Qwen3-4B: CLS 88.43% (±0.78), CWM 87.19% (±0.37), TRAJ 86.30% (±0.97). DeepSeek-Coder-6.7B shows: CLS 86.85%, CWM 86.83%, TRAJ 84.16%. The CLS–CWM gap (0.02–1.24 pp) is within seed noise in both families. However, the CLS–TRAJ gap is model-dependent: <1 pp on Qwen3 but 2.69 pp on DeepSeek-Coder, exceeding the ±2 pp equivalence margin (§3.4). CLS and CWM are robustly equivalent; TRAJ parity is model-contingent. Zero-shot baselines (Tables 1 and 3) establish that repo-level verification requires fine-tuning (0-shot→fine-tuned gains of 35–36 pp) and that formulation choice matters only at repo level (CLS outperforms CWM by 19–26 pp).

Table 1: Function-level per-test accuracy on HumanEval+MBPP (n=1,890). Mean ± std over seeds. Seeds 42/123/456: bfloat16; 789/0: NF4. DeepSeek: n=2 seeds; std reported for descriptive comparison, not inferential. TRAJ evaluated on 1,032-sample subset (447 matched variants).

Method	Model	Seeds	Accuracy (%)
Qwen3 (primary)			
CLS (fine-tuned)	4B	5	**88.43 ± 0.78**
CWM (fine-tuned)	4B	5	87.19 ± 0.37
TRAJ (fine-tuned)	4B	5	86.30 ± 0.97
DeepSeek-Coder (cross-model consistency check)			
CLS (fine-tuned)	6.7B	2	**86.85 ± 0.11**
CWM (fine-tuned)	6.7B	2	86.83 ± 1.49
TRAJ (fine-tuned)	6.7B	2	84.16 ± 1.30
Prompting baselines			
0-shot	8B (Qwen3)	1	68.70
0-shot	6.7B (DeepSeek)	1	63.00
ZS LLM-as-judge	8B (Qwen3)	1	64.40
ZS LLM-as-judge	6.7B (DeepSeek)	1	56.56
ZS LLM-as-judge	32B (NF4)	1	71.69
3-shot ICL	8B (Qwen3)	1	68.78
3-shot ICL	6.7B (DeepSeek)	1	59.95
Majority (pass)	—	—	51.75

Table 2: Model-size scaling: accuracy (%) at 4B and 8B (Qwen3, seeds 42/123, full bfloat16 precision). CLS and TRAJ scale positively; CWM does not.

Method	Size	s42	s123	Mean	Δ 4B→8B
CLS	4B	87.83	89.52	**88.68**	+1.27
CLS	8B	**90.48**	89.40	**89.94**	
CWM	4B	87.51	87.20	87.36	−0.03
CWM	8B	87.67	86.98	87.33	
TRAJ	4B	85.66	85.47	85.57	+2.13
TRAJ	8B	87.60	87.79	87.70	

3.3 Scaling Behavior

Beyond overall accuracy, we examine whether formulation ranking is stable across model sizes. Table 2 compares all three formulations at 4B and 8B using seeds 42 and 123 in full bfloat16 precision (excluding NF4 seeds to isolate the size effect).

CLS and TRAJ both scale positively from 4B to 8B (+1.27 pp and +2.13 pp respectively), while CWM remains flat (−0.03 pp). At 8B, the ranking is CLS (89.94) > TRAJ (87.70) > CWM (87.33).

The CLS–CWM gap widens from 1.32 pp at 4B to 2.62 pp at 8B (Figure 2); clustered bootstrap on the 3-seed superset (42, 123, 456 for CLS/CWM) yields $p = 0.0066$ (95% CI $[1.80, 2.71]$), which confirms the widening is not a sampling artifact.

CWM flat scaling (-0.03 pp) may reflect suboptimal hyperparameter transfer at 8B rather than a fundamental limitation; we do not claim generation *cannot* benefit from scale. At function level, outputs are short (integers, booleans) and format compliance is near-perfect; at repo level, output complexity increases substantially, and the same generation bottleneck manifests as format collapse patterns we examine in §3.5. TRAJ also shows no scaling benefit at repo level (72.92% at both 4B and 8B); only CLS scales positively across both complexity levels.

3.4 Per-Test vs. Trajectory-Level Equivalence

We compare CLS (aggregated via all-pass rule) against TRAJ on 523 variant-level instances across five seeds.[3] The pooled gap is -0.45 pp (95% CI [−1.71, +0.84]); the TOST equivalence test confirms statistical equivalence at $p = 0.007$ (± 2 pp margin, Holm-Bonferroni corrected). Our ± 2 pp threshold follows the SESOI framework (Lakens, 2017), set independently of the observed gap and grounded in deployment relevance: at function-level accuracy (85–88%), a 2 pp difference corresponds to ∼38 per-test predictions across 1,890 tests (∼2–3 problems out of 114)—comparable to between-seed variation (0.37–0.97 pp std, Table 1) and checkpoint selection noise (<1 pp, §3.7). A difference within this range is indistinguishable from training noise for pipeline selection decisions. Equivalence holds at margins as tight as ± 1.52 pp (Table 13), well below the pre-specified threshold. Within-problem predictions are correlated (intraclass correlation ICC $= 0.191$, $p < 0.0001$, design effect (DEFF) $= 3.97$); all statistical tests use clustered bootstrap accordingly. On DeepSeek-Coder, the observed CLS–CWM gap of 0.02 pp is consistent with equivalence, but with only 2 seeds the comparison is underpowered. The DeepSeek CLS–TRAJ gap (2.69 pp) exceeds the ± 2 pp margin; TRAJ parity is model-dependent rather than universal, while the CLS–CWM equivalence is robust across both families. Per-test classification provides diagnostic information (which specific tests fail) at no cost to variant-level accuracy. This parallels process supervision (Lightman et al., 2023): step-level verification enables error localization without accuracy loss.

[3] Seeds 789 and 0 use NF4 quantized inference; the remaining three use bfloat16. Restricting to bfloat16 seeds only (3 seeds) yields p=0.128: consistent in direction but underpowered due to reduced seed coverage.

Table 3: Repo-level cross-formulation comparison on SWE-bench Verified test set. Qwen3-8B CLS/TRAJ: mean ± std over 3 seeds; DeepSeek CLS: 2 seeds‡; DeepSeek TRAJ: 3 seeds; Qwen3-4B: seed 42. CWM multi-seed runs in progress.

Model	CLS (%)	CLS Var. (%)	CWM (%)	TRAJ Var. (%)◦
Qwen3-4B	83.63	64.58	58.04	72.92
Qwen3-8B	**84.83 ± 0.67**	72.92 / 81.25†	TBD	72.91 ± 3.61
DeepSeek-6.7B	84.38 ± 2.50‡	81.25†	TBD	77.08 ± 3.61
0-shot (Qwen3-8B)	49.90	—	—	—
0-shot (DeepSeek)	48.54	—	—	—
Majority	72.96	54.17	72.96	54.17

†Individual-seed variant accuracy; multi-seed aggregate pending. ‡One seed excluded: truncate preprocessing→NaN val loss, decision collapse (Table 4); mean over two matched seeds. ◦Var. columns: variant-level accuracy (n=48); CLS aggregated via all-pass rule; TRAJ by design. Per-test: n=9,784. CWM unparseable: 54% (4B); default to pass; parseable-only 38–43%, below majority. Eval seq_length differs across models (Appendix N). CWM uses epoch-3 checkpoint (val loss undefined); see text for checkpoint sensitivity analysis.

3.5 Repo-Level: Complexity-Dependent Divergence

Table 3 compares formulations at repository-level complexity. Variant-level comparisons rely on limited seed coverage; the per-test interaction effect is statistically confirmed (see below). Checkpoints are val-selected where validation loss is computable (§3.1); the test set has zero problem overlap with validation.

Zero-shot prompting collapses to an always-predict-unresolved baseline (variant accuracy 45.83%, below majority 54.17%; Appendix B).

Formulation divergence (single-seed). Under our 4–8B LoRA protocol, CLS and CWM diverge on Qwen3: the gap is 25.59 pp at 4B (CLS 83.63% vs. CWM 58.04%, seed 42; Figure 2) and 19.31 pp at 8B (84.48% vs. 65.17%). CWM falls below majority baseline at both scales. These gaps suggest complexity-dependent formulation effectiveness, though single-seed design and multiple confounds (§3.7) preclude definitive causal attribution.

Cross-model consistency. DeepSeek-Coder shows the same direction (CLS > CWM) under single-seed constraints. DeepSeek CLS degradation ranges from ∼4 pp (s42, drop preprocessing) to ∼12 pp (s123, truncate preprocessing), re-

Table 4: Repo-level failure mode taxonomy. Variant-level majority baseline: 54.17%.

Failure Mode	Representative	Fail Recall	Variant Acc
Format collapse	CWM (all)	N/A	48–52%
Decision collapse	CLS (DeepSeek, s123)	∼14%	56%
Aggr. penalty	CLS (Qwen3-8B)	51–57%	73–81%
Aggr. penalty	CLS (DeepSeek, s42)	38–51%	79–81%

flecting preprocessing and checkpoint selection sensitivity (Table 4). The CLS–CWM gap magnitude also varies across families (20 pp for Qwen3 vs. 23–28 pp for DeepSeek), primarily reflecting checkpoint selection: DeepSeek CWM's val-selected checkpoint (56.70%) underperforms its epoch-3 checkpoint (61.49%), widening the apparent gap. Under matched checkpoint selection (epoch 3 for both), the cross-model gap difference narrows to ∼2 pp (Qwen3: 20.47 pp vs. DeepSeek: 22.89 pp). The gap is formulation-driven rather than model-specific. Eval seq_length differs between models (Appendix N).

Failure mode taxonomy. Variant-level analysis (n=48, Appendix Table 10) reveals three failure modes (Table 4): format collapse (CWM, 51–74% unparseable), decision collapse (CLS under truncate preprocessing, fail recall ∼14%), and aggregation penalty (CLS under drop preprocessing, fail recall 38–57%). Decision collapse is configuration-dependent, not model-inherent. TRAJ avoids aggregation penalty by predicting variant outcomes directly (72.91–77.08%). Downsampling CLS training data to match TRAJ's data volume yields 80.74%, still 7.82 pp above TRAJ; this controls data volume but not label granularity (per-test vs. variant-level), and remains a single-seed comparison (Appendix B).

Default assignment artifact. The apparent CWM accuracy increase from 4B to 8B is an artifact of rising unparseable rate (53.7%→73.8%), which inflates accuracy via majority-class default while parseable-only accuracy decreases (42.88%→38.52%; Appendix M).

Complexity as moderator. The CLS–CWM gap widens from 1–3 pp (function) to 19–26 pp (repo) under our protocol; a matched-context ablation confirms input disparity explains <3% of this gap (Appendix B).

Formulation×complexity interaction. A mixed-effects linear probability model with problem-level random intercepts (n=56,480 observations, 139 problem clusters, DeepSeek-Coder) confirms the interaction: the CWM effect at function level is negligible ($\hat{\beta} = -0.002$, p=0.805), but the formulation×complexity interaction is highly significant ($z = -23.7$, $p<0.001$). CWM and CLS are statistically indistinguishable at function level ($\Delta = -0.2$ pp) but diverge by 22 pp at repo level, confirming the divergence is not an artifact of aggregation. The cross-complexity analysis uses DeepSeek-Coder, the only family with both function- and repo-level CWM evaluations under matched conditions. Across both model families at repo level, the CWM deficit is −22 pp ($z = -64.4$, $p<0.001$); the model family effect is significant but small (+1.1 pp for Qwen3, p=0.002), and the formulation×model interaction is likewise small (+1.6 pp, p=0.006). Formulation explains >95% of the predictable variance; model identity modulates gap magnitude without changing direction (Appendix L).

Additional repo-level limitations appear in Appendix O.

3.6 Analysis

Aggregation penalty. Under the all-pass rule, a single false negative flips the variant verdict. The penalty ranges from 1.36 pp (DeepSeek s42) to 19 pp (Qwen3-4B), depending on fail recall and test count (Appendix Table 10). Val-based checkpoint selection biases toward per-test accuracy at the expense of fail recall, which amplifies the penalty. TRAJ avoids this by predicting variant outcomes directly (72.91–77.08%). The penalty is predictable from test count: function-level variants ($\bar{T} \approx 2.3$) incur negligible penalty, while repo-level variants ($\bar{T} \gg 10$) incur substantial penalty, both the *advantage* and *cost* of per-test granularity concentrate at high T (Appendix H(a)).

Why does generation degrade at repo level? Output space complexity increases nonlinearly: function-level outputs are short strings (integers, booleans); repo-level outputs are unconstrained execution traces (mean ∼477 tokens) where errors compound multiplicatively, and cause format collapse (53–74% unparseable). Even among parseable outputs, accuracy (38–43%) is below majority baseline; forced parsing that rescues 90.6% of unparseable outputs achieves only 49.4% accuracy, confirming the bottleneck is semantic rather than syntactic (Appendix H(c)).

CLS achieves 217× higher throughput than CWM (1 output token vs. avg 477), consistent across both model families.

Error complementarity. At function level, CLS and CWM exhibit partial complementarity: both are FP-dominant (over-predict failure), but at different rates (CLS 56.5%, CWM 70.3%; Appendix H(a)), so CWM over-predicts failure on instances CLS correctly classifies. An oracle combination achieves 90.63% (+2.80 pp over the best single formulation; single-seed estimate). At repo level, oracle routing across configurations reaches 95.83% variant accuracy (+14.58 pp over the best single configuration, n=48), the strongest evidence that the formulation design space contains complementary solutions worth exploring. On SWE-bench Verified (500 instances), the per-test advantage concentrates at $T \geq 2$ fail-to-pass tests (31.3% of instances, contributing 55.3% of the total oracle advantage; Appendix A). A re-repair pilot found no benefit from per-test diagnostics (n=44, p=0.42; Appendix H(j)). Extended analysis appears in Appendix H.

3.7 Threats to Validity

We identify five potential confounds and assess each in Appendix N: checkpoint selection (<1 pp variation in ablation; NaN validation loss forces final-epoch default in all CWM runs and DeepSeek CLS s123), evaluation sequence length (∼1.5 pp), preprocessing (∼8 pp cross-configuration variance), TRAJ training regime (downsample confirms formulation advantage), and CWM scale/pretraining (primary limitation; all repo-level CWM conclusions conditional on the ≤8B LoRA regime). The 19–26 pp gap exceeds any single assessed confound, but correlated confounds and limited seeds mean definitive causal attribution requires multi-seed replication. Holm-Bonferroni correction confirms the two core claims survive (adjusted p=0.012 and p=0.021; Table 15).

4 Related Work

Execution Simulation and Verification. Execution trace generation improves code reasoning (Nye et al., 2021), with extensions to output prediction (FAIR CodeGen team et al., 2025; Maimon et al., 2026; Liu et al., 2024, 2023; Lee and Huang, 2026), failure analysis (Abdollahi et al., 2025), and error detection (Li et al., 2024). Verification formulations span trajectory-level reward (Shum et al., 2025; Zhu et al., 2026), process-level feedback (Li et al., 2025b; Xu et al., 2025), per-test classification (Ni et al., 2023; Chen et al., 2023), learned metrics (Zhuo, 2024; Dong et al., 2024; Zhou et al., 2023), and RL with execution-based rewards (Wang et al., 2026; Tang et al., 2026). Aletheia (Venkatkrishna et al., 2026) compares training recipes; we compare *formulations*. Test-time scaling (Snell et al., 2025; Li et al., 2025a; Chen et al., 2025) and self-consistency (Wang et al., 2023) demonstrate verification value; none compare formulations at matched capacity.

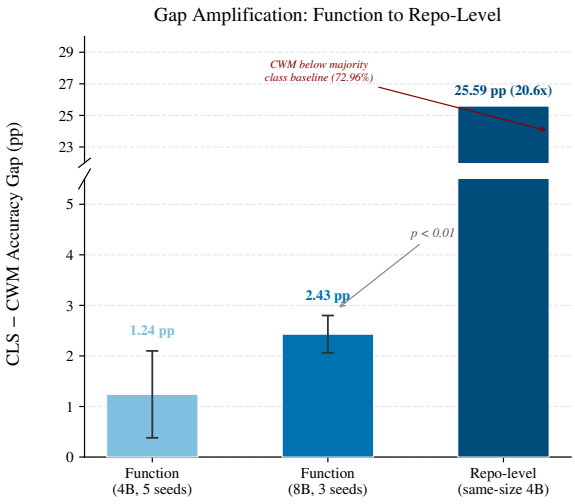


Figure 2: Gap amplification from function to repo level. At function level, the CLS–CWM gap is 1–3 pp; at repo level, it widens to 19–26 pp with CWM falling below majority baseline.

Process vs. Outcome Supervision. PRMs outperform ORMs for math reasoning (Lightman et al., 2023; Uesato et al., 2022), with automated annotation (Wang et al., 2024; Luo et al., 2024) and code-specific extensions (Zhang et al., 2025; Ye et al., 2025; Zhang et al., 2026; Jiang et al., 2025). Our CLS provides per-test (process) supervision while TRAJ provides outcome-level; the PRM literature compares supervision granularity given a fixed output format, while we additionally vary the output format (CWM generation vs. CLS classification).

Repair, Mutation Testing, and Task Framing. CodeRL (Le et al., 2022), AlphaCode (Li et al., 2022; AlphaCode Team, 2023), and Reflexion (Shinn et al., 2023) use execution feedback for repair. SWE-bench (Jimenez et al., 2024;

Chowdhury et al., 2024) provides benchmarks; SWE-agent (Yang et al., 2024) and Agentless (Xia et al., 2025) demonstrate pipelines where verification is a bottleneck (Yang et al., 2025b). Predictive mutation testing (Tufano et al., 2019; Kim et al., 2022; Wang et al., 2025; Jain et al., 2023) shares our per-test structure. LoRA (Hu et al., 2022) and QLoRA (Dettmers et al., 2023) enable parameter-efficient fine-tuning. Task framing—how a problem is cast as a model's input–output mapping—systematically affects performance: Raffel et al. (2020) showed that classification, generation, and span-extraction formulations yield different accuracy–efficiency tradeoffs across NLP tasks. We apply this paradigm to code verification, holding architecture and training data constant while varying only the formulation; our finding that complexity moderates the classification–generation tradeoff extends their analysis to a new domain.

5 Conclusion

We presented the first systematic comparison of three execution-free verification formulations (CLS, CWM, TRAJ). At function level, CLS and CWM are statistically equivalent (TOST p=0.007, 5 seeds); TRAJ reaches parity on Qwen3 but not DeepSeek-Coder (2.69 pp). Per-test classification achieves 217× higher throughput while providing diagnostic signal whose downstream utility remains unconfirmed (n=44, p=0.42). Complementary errors (oracle: 90.63%/95.83%; single-seed) motivate complexity-aware routing. At repository level under ≤8B LoRA fine-tuning, a significant interaction ($z = -23.7$) confirms divergence (19–26 pp, single-seed variant-level), with CWM falling below majority baseline—consistent across families but dependent on multiple confounds (§3.7). Variant-level analysis reveals three distinct failure modes (format collapse, decision collapse, aggregation penalty) consistent across model families. At function level, formulation choice is immaterial; at repository level under ≤8B LoRA, CLS is recommended for both accuracy and throughput. Behavior at larger scale remains open (FAIR CodeGen team et al., 2025).

Limitations

Most repo-level comparisons use limited seeds; the gap (19–26 pp) exceeds any single assessed confound (§3.7), but correlated confounds mean definitive causal attribution requires multi-seed replication. Our CWM baseline uses ≤8B LoRA fine-tuning, not the original 32B CWM with execution trace mid-training (FAIR CodeGen team et al., 2025); at larger scale, the generation bottleneck may diminish. Training data comprises synthetic mutations (Andrews et al., 2005; Just et al., 2014); the synthetic-to-real transfer gap for absolute accuracy remains untested, though the coupling effect supports relative ranking validity (Appendix O). Model coverage is limited to 6–8B code-specialized LLMs; generalization to larger or general-purpose models is untested. All code, data, and model checkpoints will be released upon acceptance.

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Responsible NLP Research Checklist

This paper follows the ARR Responsible NLP Research checklist.

A. For Every Submission

A1. Did you describe the limitations of your work? Yes. Section 5 (Limitations paragraph) discusses: single-seed repo-level comparisons, synthetic-to-real transfer gap, lack of downstream re-repair validation, and val-test mutation distribution mismatch. Section 3.5 (Limitations paragraph) discusses val-best checkpoint selection variance and single-seed constraints.

A2. Did you discuss any potential risks of your work? This work compares formulations for execution-free code verification on public benchmarks. The primary application is improving automated program repair pipelines. We identify no significant risks: the methods predict test outcomes rather than generating executable code, the datasets are derived from open-source repositories, and no personally identifiable information is involved. Misuse potential is limited because the verifiers require task-specific fine-tuning on labeled data and do not generalize to arbitrary code generation.

B. Did You Use or Create Scientific Artifacts?

B1. Did you cite the creators of artifacts you used? Yes. All datasets (HumanEval (Chen et al., 2021), MBPP (Austin et al., 2021), SWE-bench Verified (Chowdhury et al., 2024)), models (Qwen3 (Yang et al., 2025a), DeepSeek-Coder (Guo et al., 2024)), and methods (LoRA (Hu et al., 2022), QLoRA (Dettmers et al., 2023)) are cited in Sections 3.1 and 2.1.

B2. Did you discuss the license or terms for use and/or distribution of any artifacts? HumanEval is released under MIT license; MBPP under CC-BY-4.0; SWE-bench Verified under MIT license. Qwen3 models are released under Apache 2.0; DeepSeek-Coder under the DeepSeek License. All uses are consistent with their respective licenses. We do not release new artifacts in this submission.

B3. Did you discuss if your use of existing artifact(s) was consistent with their intended use? Yes. All benchmarks are used for their intended purpose of evaluating code understanding and generation capabilities. SWE-bench Verified is used for repository-level evaluation as designed by its creators.

B4. Did you discuss the steps taken to check whether the data contains any information that names or uniquely identifies individual people or offensive content? Not applicable. Our datasets consist of code snippets and test cases from open-source repositories. The mutation-based data construction process operates on source code only and does not involve personal data or natural language content that could contain offensive material.

B5. Did you provide documentation of the artifacts? Yes. Section 3.1 documents dataset statistics (sizes, splits, label distributions, majority baselines). The mutation-based data construction process is described in Section 3.1. Appendix E provides full hyperparameter details.

B6. Did you report relevant statistics like the number of examples, details of train/test/dev splits, etc.? Yes. Section 3.1: function-level 19,076 samples (train 15,290 / val 1,896 / test 1,890; pass:fail ≈ 52:48); repo-level ∼70K training, val n=9,335, test n=9,784 (majority baseline 72.96%).

C. Did You Run Computational Experiments?

C1. Did you report the number of parameters in the models used, the total computational budget, and computing infrastructure? Yes. Models: Qwen3-4B (∼4B parameters, LoRA rank 16 ≈ 4M trainable), Qwen3-8B (∼8B), DeepSeek-Coder-6.7B (∼6.7B). In-

frastructure: 4× NVIDIA RTX 5090 GPUs. Total compute: ∼370 GPU-hours across 223 completed runs (∼46 configurations including seeds, formulations, model sizes, and ablations). Appendix E provides per-run details.

C2. Did you discuss the experimental setup, including hyperparameter search and best-found hyperparameter values? Yes. Section 3.1 reports the training configuration (LoRA rank 16, 3 epochs, 5 seeds with mixed precision). Appendix E provides full hyperparameter values. We use a fixed configuration rather than extensive hyperparameter search; repo-level models use validation accuracy for checkpoint selection.

C3. Did you report descriptive statistics about your results with, e.g., error bars or confidence intervals? Yes. All multi-seed results report mean ± standard deviation. Confidence intervals use clustered bootstrap (114 problem clusters, 10K resamples; Section 3.1). Equivalence testing uses TOST ($p = 0.007$, ±2 pp margin; Section 3.4). ICC quantifies within-problem correlation with design effect reported.

C4. If you used existing packages, did you report the implementation and version numbers? Yes. Implementation uses HuggingFace Transformers (≥4.44), PEFT (≥0.12), PyTorch (≥2.0), bitsandbytes (≥0.43, for NF4 quantization), and Python 3.10. LoRA (Hu et al., 2022) and QLoRA (Dettmers et al., 2023) configurations are detailed in Appendix E. Exact version pinning is provided in the code repository.

D. Did You Use Human Annotators?

Not applicable. This work uses no human annotators or participants. All labels are derived from automated test execution.

E. AI Assistants

E1. Did you use AI assistants (e.g., ChatGPT, Copilot) in any part of the paper writing or coding process? Yes. AI coding assistants were used during development of training and evaluation scripts. AI writing assistants were used for drafting and editing portions of the manuscript. All scientific claims, experimental design, and analysis were conducted and verified by the authors.

A Oracle Accuracy-Utility Analysis

We stratify SWE-bench Verified instances by F2P test count T, comparing per-test vs. trajectory-level oracle resolve rates under i.i.d. and block-correlated error models (ICC = 0.191, from function-level analysis; §3.4).

Table 5: Oracle accuracy-utility frontier: per-test advantage stratified by F2P test count (T) on SWE-bench Verified (500 instances). Block-correlated model uses function-level ICC = 0.191 (from function-level analysis, §3.4). P2P regression rate: 1.04% mean, 0% median.

T	Inst.%	i.i.d. (pp)	Block (pp)	Contrib.%
1	68.7	−0.04	−0.09	≈0
2	19.1	18.49	31.69	55.3
3–5	7.1	25.00	40.74	≈26
≥ 6	5.1	30.00	40.06	≈19
All	100.0	—	**10.93**	100.0

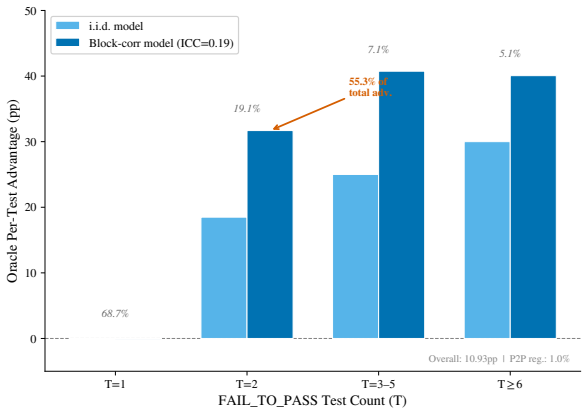


Figure 3: Oracle per-test advantage by F2P test count T. Advantage is zero at T=1 (68.7%) and concentrates at T=2 (19.1%, contributing 55.3% of total advantage).

Table 5 and Figure 3 reveal a sharp concentration: at $T = 1$ (68.7% of instances), per-test and trajectory-level are informationally equivalent. The advantage materializes at $T = 2$ (19.1%), where block-correlated advantage reaches 31.69 pp (55.3% of total). The overall block-correlated advantage is 10.93 pp, driven by the $T \geq 2$ minority. Block correlation amplifies relative to i.i.d.: at $T = 2$, 31.69 pp vs. 18.49 pp (1.7×). P2P regression is negligible (1.04% mean, 0% median); the practical value concentrates in F2P discrimination.

Sensitivity to ICC estimate. The analysis above uses ICC = 0.191 from function-level analy-

sis (§3.4), where mean cluster size $\bar{k} \approx 16.5$ yields DEFF $= 3.97$. At repository level, within-problem correlation is substantially stronger: ICC $= 0.3146$ with DEFF $= 114.4$ (§3.5), a consequence of the larger and more heterogeneous test suites in SWE-bench Verified problems. Under higher ICC, within-problem errors cluster more tightly and amplify the block-correlated per-test advantage at each T stratum; the correction from i.i.d. to block-correlated grows monotonically with ICC. The function-level ICC $= 0.191$ used in Table 5 thus provides a conservative lower bound on the per-test advantage. The repo-level correlation structure would yield an even larger gap between per-test and trajectory-level oracle resolve rates, which strengthens the case for per-test granularity at repository-level complexity.

B Ablation Studies

LoRA rank sensitivity. All CWM experiments use LoRA rank r=16. To test whether this capacity constraint handicaps CWM, we train rank-32 variants (seed 42), doubling adapter parameters.

Table 6: LoRA rank ablation for CWM 4B (seed 42, 3 epochs, lr $= 2\times10^{-4}$). Rank-16 seed variance: 0.53 pp (seeds 42 vs. 456).

Config	α/r	Acc (%)	Fail F_1	Pass F_1	Δ
r=16, α=32	2	**87.51**	—	—	—
r=32, α=32	1	87.00	87.1	86.9	−0.51
r=32, α=64	2	87.04	87.27	86.79	−0.47

Both rank-32 variants fall within the 2-seed variance of the rank-16 baseline (0.53 pp between seeds 42 and 456). The α=64 variant (matching the baseline α/r=2 ratio) rules out the confound that the α=32 result might reflect reduced effective scaling rather than rank insensitivity. This is consistent with CWM's function-level performance being bounded by output-format constraints rather than adapter capacity.

Majority voting. Majority voting (K=3) decreases accuracy from 87.51% to 86.83% (−0.68 pp): errors are systematic rather than stochastic, so sampling-based ensembles do not help.

Sequence-length ablation details. Both formulations are trained with `max_seq_length`=2,048. At inference, CWM reserves `max_new_tokens`=512, leaving ∼1,536 effective input tokens; CLS allocates nearly all context to input (5,120–8,192 tokens). A matched-context ablation (CLS 8B at 2,048 tokens) achieves 83.04%, only 1.44 pp below the 8,192-token evaluation (84.48%), though truncation introduces 136 unparseable outputs (1.39% of test samples). The matched-context gap (CLS@2,048 vs. CWM@2,048) is 25.00 pp (98% of the full 25.59 pp gap), confirming that input context disparity explains at most 3% of the difference.

Training data size control. To control for the training set size asymmetry between CLS (∼70K instances) and TRAJ (386 variant-level instances, ∼8K per-test equivalents, at repo level), we downsample CLS training data to match TRAJ in data volume. Downsampled CLS (Qwen3-4B, seed 42) achieves 80.74% repo-level per-test accuracy, only 2.89 pp below the full-data CLS (83.63%) and still 7.82 pp above TRAJ (72.92%). The CLS formulation advantage is not attributable to training data volume.

Few-shot prompting baselines. We evaluate 3-shot in-context learning at repo level to test whether prompting strategy closes the gap to fine-tuned models. Table 7 reports both per-test and variant-level accuracy.

Table 7: Repo-level prompting baselines (per-test n=9,784; variant-level n=48). Majority baseline: 72.96% per-test, 54.17% variant-level.

Model	0-shot (per-test)	3-shot (per-test)	3-shot (variant)
Qwen3-8B	49.90%	71.61%	45.83%*
DeepSeek-6.7B	48.54%	46.28%	45.83%*

*Both models achieve 45.83% by correctly classifying all 22 unresolved variants and zero resolved variants (22/48 = 45.83%), equivalent to an always-predict-unresolved baseline. The 100% overlap in variant-level predictions confirms complete degradation at task-relevant granularity.

Both models converge to exactly 45.83% variant-level accuracy under 3-shot prompting, below majority baseline (54.17%). Variant-level analysis reveals this is not a coincidence: both models correctly classify all 22 unresolved variants and zero resolved variants, equivalent to an always-predict-unresolved baseline with 100% prediction overlap. Per-test accuracy diverges sharply (Qwen3-8B 71.61% vs. DeepSeek 46.28%), but this divergence is entirely superficial: Qwen3-8B predicts pass on nearly all tests, inflating per-test accuracy on a pass-heavy dataset

(72.96% pass rate) without genuine discriminative ability. Per-test metrics can thus severely misrepresent baseline capability; variant-level evaluation is necessary to detect complete degradation. Neither 0-shot nor few-shot prompting approaches fine-tuned performance (82–84% per-test); fine-tuning is a necessary condition for repo-level verification.

C Repo-Level Per-Mutation Breakdown

Table 8 compares per-mutation accuracy at repo level. CLS produces 0% unparseable outputs; all values are direct classification accuracy. CWM overall accuracy includes default-pass assignment for unparseable outputs (see §3.5).

Table 8: Per-mutation test accuracy (%) at repo level (SWE-bench Verified, n=9,784). CLS 8B reports two seeds; Δ is mean 4B→8B gain for CLS. CWM 4B per-mutation overall accuracy omitted for non-gold mutations due to low parse rates (4.4–56.9%, Table 9); gold overall accuracy derived from parse rate 43.5% and parseable accuracy 23.5%.

Mutation	n	CLS 4B	CLS 8B s42	CLS 8B s123	CWM 4B	CWM 8B	Δ CLS
gold	4,633	98.25	96.24	99.60	66.72	80.06	−0.33
operator_swap	3,357	78.91	81.38	82.60	—	54.57	+3.08
off_by_one	183	58.47	86.89	79.20	—	63.39	**+24.58**
return_value	562	67.97	75.44	66.40	—	49.11	+2.95
boundary_removal	1,049	46.90	46.90	44.90	—	42.23	−1.00

Table 9 shows that CWM generation quality degrades from 4B to 8B across all mutation types.

Table 9: CWM parse rates and parseable-only accuracy per mutation type (seed 42). Parse rate decreases uniformly from 4B to 8B; parseable-only accuracy is below random binary chance (50%) for gold at both scales.

	Parse Rate (%)		Parse Acc (%)	
Mutation	**4B**	**8B**	**4B**	**8B**
gold	43.5	23.9	23.5	16.5
operator_swap	56.9	21.2	63.1	58.8
off_by_one	4.4	27.3	0.0	38.0
return_value	48.8	26.9	27.0	56.9
boundary_removal	30.6	51.8	59.2	51.8

Key findings: (1) Gold parse rate collapses from 43.5% to 23.9% at 8B; the simplest generation target (unmodified code, where all tests pass) degrades most, consistent with a fundamental generation bottleneck rather than task difficulty. (2) Among parseable outputs, gold accuracy is 16.5–23.5%: the model generates outputs that fail exact-match even for correct code. (3) `off_by_one` is an anomaly: parse rate *increases* from 4.4% to 27.3%, but this is from an extremely low base (n=8 parseable at 4B), making both directions unreliable for this category.

D Variant-Level Accuracy

Table 10 reports variant-level accuracy (n=48) alongside per-test accuracy for all repo-level configurations. For CLS, the aggregation penalty (per-test → variant-level drop) ranges from 1 pp to 19 pp and is largest for models with low fail recall under the all-pass rule. CWM variant-level accuracy (48–52%) falls at or below the majority baseline, consistent with format collapse as the dominant failure mode.

Table 10: Repo-level per-test vs. variant-level accuracy (n=48 variants). Penalty = per-test − variant-level. TRAJ predicts variant outcomes directly (no aggregation). Variant-level majority baseline: 54.17% (26/48).

Model	Seed	Per-test (%)	Variant (%)	Penalty (pp)	Fail Recall (%)
CLS					
Qwen3-8B	s42	84.48	72.92	11.56	56.76
Qwen3-8B	s123	85.60	81.25	4.35	51.44
Qwen3-4B	s42	83.63	64.58	19.05	48.72
DeepSeek-6.7B	s42 (e1)	82.61	81.25	1.36	51.02
DeepSeek-6.7B	s42 (e3)	82.51	79.17	3.34	38.47
DeepSeek-6.7B	s123	74.93	56.25	18.68	∼13.72
CWM					
Qwen3-8B	s42	65.17	52.08	13.09	—
DeepSeek-6.7B†	s123	—	47.92	—	—
TRAJ					
Qwen3-4B	s42	—	72.92	—	—
Qwen3-8B	s42	—	70.83	—	—
DeepSeek-6.7B	3-seed	—	77.08 ± 3.61	—	—
Majority	—	72.96	54.17	—	—

†Evaluated on ∼75% of test instances (7210–7668 of 9784). DeepSeek TRAJ Table 3 reports 3 seeds (42/123/456) for alignment with CLS/CWM seed coverage; three supplementary seeds (0/789/999) yield 75.0%, 66.7%, 79.2% respectively (6-seed mean 75.35 ± 5.00%), consistent in direction.

E Training Hyperparameters

Table 11 lists all hyperparameters used for fine-tuning. Both CLS and CWM share the same LoRA configuration, optimizer, and training schedule; the only differences are per-device batch size, gradient accumulation steps (adjusted to maintain identical effective batch size of 32 across model sizes), and learning rate (DeepSeek-Coder uses 2×10^{-4} vs. 5×10^{-5} for Qwen3, selected via grid search over $\{5\times10^{-5}, 1\times10^{-4}, 2\times10^{-4}\}$ on held-out validation loss).

Data processing. DeepSeek CWM seeds use `train_unified.py`, which drops samples exceeding `max_seq_length` (3,405 of 70,748 samples, 4.8%). Qwen CWM seeds use `train_repo_cwm.py`, which truncates and retains all samples. This difference affects to-

Table 11: Training hyperparameters for CLS/CWM configurations. TRAJ models use lr $= 2\times10^{-4}$ due to the smaller training set (∼8K function-level instances vs. ∼70K for CLS/CWM); other hyperparameters are shared.

Shared across all configurations	
LoRA rank / alpha / dropout	16 / 32 / 0.05
LoRA targets	`q/k/v/o_proj`
Optimizer	AdamW (β_1=0.9, β_2=0.999)
LR scheduler	Linear
Effective batch size	32
Max sequence length	2,048
Epochs	3
Precision	bf16
Gradient checkpointing	✓
Seeds (function)	{42, 123, 456, 789, 0}[b]
Seeds (repo)	{42, 123, 456}[c]

Model-family–specific parameters			
	Qwen3-4B	**Qwen3-8B**	**DS-6.7B**
Learning rate	5e-5	5e-5	2e-4
Weight decay	0.01	0.01	0.0
Warmup ratio	0.05	0.05	0.1
Batch / grad. accum. (CLS)	2 / 16	1 / 32	2 / 16
Batch / grad. accum. (CWM)	1 / 32	1 / 32	1 / 32
Training samples	70,748	70,748	67,343–70,748[a]

[a]DeepSeek CLS canonical run (s42) uses drop preprocessing, which excludes 3,405 sequences exceeding max length (67,343 of 70,748). The s123 run uses truncate preprocessing (70,748 samples) but produces NaN validation loss. See §3.1. [b]Seeds 789 and 0 use NF4 quantization; others use bf16. [c]Repo-level seed coverage varies by configuration; see Table 3 caption for per-model counts.

tal training steps (∼12,627 vs. ∼13,266 for DeepSeek, ∼6,633 for Qwen) but not the training data distribution, as dropped samples are uniformly distributed across labels.

Prompt templates. Prompt templates for all formulations (CLS, CWM, TRAJ) are provided in the code repository. CLS uses a binary classification template ("Does this test pass or fail?"); CWM uses an execution simulation template ("What is the output of this test?"); TRAJ uses a variant-level verdict template ("Is this patch resolved or unresolved?").

Mutation operators. We use five mutation types: (1) *gold*: unmodified code (all tests pass); (2) *operator_swap*: replaces comparison or arithmetic operators (e.g., < → <=); (3) *off_by_one*: shifts loop bounds or array indices by ±1; (4) *return_value*: modifies return statements (e.g., negating a boolean, changing a constant); (5) *boundary_removal*: removes conditional guards or boundary checks. Mutation operators were applied using AST-level transformation; variants that did not alter any test outcome were discarded.

Compute budget. All experiments were conducted on NVIDIA RTX 5090 GPUs. Training time is approximately 1.5 hours per run for 4B models and 2 hours for 8B models. Total compute budget: ∼370 GPU-hours across 223 completed runs (135 successful, 88 failed/killed; ∼46 unique configurations including seeds, formulations, model sizes, and ablations).

F Val-Test Mutation Type Distribution

Table 12 reports the mutation type composition of the repo-level validation and test sets. The most notable shift is `operator_swap`, which nearly doubles from val (19.1%) to test (34.3%), while `off_by_one` shrinks from 10.0% to 1.9%.

Table 12: Repo-level mutation type distribution: validation vs. test set, with per-type CLS accuracy at two model sizes (seed 42). Shift = test% − val%.

Mutation Type	**Val (%)**	**Test (%)**	**Shift**	**CLS 4B**	**CLS 8B**
gold	45.2	47.4	+2.2	98.3	96.2
operator_swap	19.1	34.3	+15.2	78.9	81.4
boundary_removal	15.0	10.7	−4.3	46.9	46.9
return_value	10.7	5.7	−5.0	68.0	75.4
off_by_one	10.0	1.9	−8.1	58.5	86.9

The `operator_swap` shift is the primary source of val-test distribution mismatch. Since CLS achieves moderate accuracy on this type (78.9–81.4%), the over-representation in the test set does not systematically bias overall accuracy estimates in either direction. Two boundary observations are notable: (1) `boundary_removal` represents a scale-invariant difficulty ceiling at ∼47% across both model sizes: this mutation type resists capacity scaling; (2) `off_by_one` shows the largest scaling gain (+28.4 pp, 58.5%→86.9%) but its small test share (1.9%) limits its contribution to overall accuracy differences across scales.

G Qualitative Error Analysis

To complement the quantitative divergence reported in §3.5, we examine representative errors from each formulation at repository level.

CWM failure modes. Two patterns account for the majority of unparseable CWM outputs on Qwen3. (1) *Task misunderstanding*: the model generates a complete simulated test runner log (test names, timing, summary lines) rather than a pass/fail verdict. For instance, on `django__django-14315`, the model outputs a fabricated 1,248-token Django test exe-

cution trace with hallucinated test names that do not exist in the target file. The model confuses "predict test outcome" with "simulate test execution environment." (2) *Autoregressive degeneration*: the model enters a repetitive loop (e.g., repeating `from _pytest.config import _parse_pytest_args` until truncation at 4,542 tokens). Both patterns produce zero usable signal and default to the majority-class label. This explains why unparseable predictions carry higher accuracy than parseable ones (§3.5).

CLS failure modes. CLS errors concentrate in two categories requiring multi-step reasoning. False positives arise from *cross-concern value propagation*: a boundary removal mutation in `pytest` converts a guarded attribute access (`if xfailed and not xfailed.run`) to `if True:`, causing a NoneType error on tests exercising unrelated skip marks. The model fails to trace the None path through the pytest hook lifecycle. False negatives arise from *mathematical equivalence*: on `HumanEval/130` (Tribonacci variant), the model predicts failure for correct code because verifying the self-referential recurrence requires algebraic substitution rather than syntactic inspection. This single problem accounts for 17% of all function-level false negatives.

Complementarity. These failure patterns operate at different levels: CWM fails at the format level (producing no usable signal), while CLS fails at the reasoning level (producing a confident but incorrect binary verdict). Both formulations are FP-dominant (over-predict failure), but CWM's higher FP rate (70.3% vs. 56.5%) means it over-predicts failure on instances CLS correctly classifies, producing partial complementarity (oracle combination: 90.63%; §H, paragraph a).

H Extended Analysis

(a) Error complementarity. CLS and CWM are both FP-dominant at function level but at different rates, producing partial complementarity. Among CLS errors (seed 42), 56.5% are false positives (PASS → FAIL); among CWM errors, 70.3% are false positives. Both formulations over-predict failure, but CWM's stronger FP bias means it incorrectly rejects passing tests that CLS correctly accepts. An oracle combination achieves 90.63% accuracy (1713/1890 correct, 177 both-wrong; +2.80 pp over the best single formulation, CLS 87.83%; single-seed estimate, seed 42). The complementarity arises from different *degrees* of fail-bias rather than opposite error directions: CWM's generation bottleneck produces aggressive failure predictions, while CLS's single-token output is more calibrated. Ensemble approaches exploiting this degree difference are a natural direction.

(b) CWM error mode breakdown. Among unparseable CWM outputs (Qwen3, repo-level), 76.9% contain no execution keywords; 22.9% contain ambiguous co-occurring pass/fail keywords. Format compliance varies sharply across repositories (scikit-learn: near-zero unparseable; sphinx: 100%); repository-specific code patterns interact with the model's generation capabilities in complex ways. This per-repository variation is consistent with a model/tokenizer-specific failure rather than a formulation-level one.

(c) Relaxed matching does not rescue CWM. Relaxed matching (tolerating format deviations, L0–L4) *decreases* CWM accuracy: 58.04%→56.98% (seed 42) and 59.09%→55.91% (seed 123). Unparseable predictions default to the majority class (pass, 72.96%), which outperforms the model's own parseable predictions (38–43%). The divergence widens from 26 pp to 27 pp under relaxed matching. This rules out the hypothesis that parsing strictness inflates the observed gap. Constrained decoding (Willard and Louf, 2023) could enforce format compliance but would not address the semantic accuracy deficit among parseable outputs.

(d) Variance compression. CWM's 5-seed standard deviation (0.37 pp) is lower than CLS (0.78 pp) and TRAJ (0.97 pp) at function level. If CWM performance is capped by output-format constraints at a given complexity level, seed-to-seed variability is compressed relative to formulations whose performance depends more on learned representations.

(e) Per-test precision and recall. CLS achieves fail F_1 87.59% (precision 86.20%, recall 89.04%) at seed 42.Multi-F2P instances (6–10 tests) reach 96.3% accuracy, consistent with the oracle finding that per-test granularity is most valuable for complex instances.

(f) Block correlation. Within-problem errors are correlated (ICC = 0.191, DEFF = 3.97,

2.81× more all-correct problems than i.i.d. expectation). All statistical tests use clustered bootstrap accordingly.

(g) Failed interventions. We systematically tested four interventions targeting different hypothesized causes of CWM repo-level degradation; all failed:

- **Capacity (rank 32):** −0.51 pp, within seed variance (§B).
- **Majority voting (K=3):** −0.68 pp; 93.3% unanimous, net flips negative (§B).
- **Relaxed matching (L0–L4):** −1.06 pp; default-pass outperforms parseable predictions (paragraph (c) above).
- **Model scaling (4B→8B):** parseable accuracy decreases (42.88%→38.52%) while unparseable rate rises (53.7%→73.8%); gain is a default-assignment artifact (§3.5).

Each intervention targets a distinct alternative explanation (insufficient capacity, random noise, parsing strictness, model scale); the uniform failure across all four confirms the bottleneck is formulation-level—output space complexity at repo level exceeds what ≤8B generation models can handle, independent of these surface-level factors.

(h) Per-mutation scaling patterns. Scaling benefits are highly non-uniform across mutation types (Table 8). `off_by_one` shows the largest CLS scaling gain: 58.5% (4B) to 86.9% (8B, +28.4 pp). In contrast, `boundary_removal` is scale-invariant at ∼47% across both sizes, which marks an intrinsic difficulty ceiling. For CWM, gold mutation parse rate collapses from 43.5% (4B) to 23.9% (8B) despite being the simplest target: generation capability degrades even for trivially correct code at larger scale within Qwen3. The val-test mutation type distribution differs notably (`operator_swap` shifts from 19.1% to 34.3%; Table 12), but this shift does not systematically bias overall accuracy because CLS performance on `operator_swap` (78.9–81.4%) is close to the overall mean.

(i) Inference efficiency. CLS achieves 217× higher throughput than CWM at repo level (853 vs. 3.9 inferences/min; Qwen3-4B, RTX 5090, 100 samples, avg ∼1,040 input tokens). The speedup derives directly from the formulation: classification requires a single output token, whereas CWM generates an average of 477 tokens per inference. Median latency is 48ms (CLS) vs. 16.3s (CWM). At the observed unparseable rate of 73.8% (8B), the majority of CWM's generation budget produces tokens that cannot yield a valid verdict. This CLS–CWM efficiency gap is an inherent property of the formulation difference (1 vs. $O(L)$ output tokens) and should generalize across models, unlike the accuracy divergence which may be model-contingent.

The CLS–TRAJ efficiency comparison is more nuanced, because CLS makes one call per test while TRAJ makes one call per instance. Per-call, CLS is 20–138× faster (depending on input length: 47ms for short contexts vs. 2,632ms for TRAJ). However, the per-instance effective speedup depends on test count $|T|$: at function level ($|T| \approx 2.3$), CLS is ∼24× faster (108ms vs. 2,632ms per instance); at repo level ($|T| \approx 12.3$, ∼70ms/call with longer inputs), CLS is ∼3× faster (861ms vs. 2,631ms). A crossover exists at $|T| \approx 50$ tests, beyond which TRAJ's single-call amortization dominates. This boundary condition is informative for deployment: CLS is strongly preferred for function-level and moderate repo-level test suites, while extremely large test suites (>50 tests per instance) favor trajectory-level efficiency despite its coarser diagnostic signal.

(j) Re-repair pilot. We tested whether per-test diagnostic feedback improves downstream repair. A Qwen3-8B repair agent received either per-test diagnostic signals (which specific tests failed and why) or binary accept/reject feedback, evaluated on 44 SWE-bench Verified instances. The diagnostic condition achieved accuracy no higher than the binary baseline ($A_1 = 52.3\%$, $p = 0.42$, effect size ≈ 0). Per-test diagnostic information does not automatically translate to repair improvements at this scale. The negative result may reflect insufficient agent capability to exploit fine-grained feedback, insufficient sample size (at $n = 44$, statistical power is ∼0.3 for a medium effect size), or a genuine limitation of the diagnostic-to-repair translation.

I TOST Margin Sensitivity

Table 13 reports the TOST equivalence p-value as a function of the equivalence margin, computed on the 5-seed variant-level CLS–TRAJ comparison (gap = −0.45 pp, 523 variants, 114 clusters, clustered bootstrap).

Table 13: TOST margin sensitivity for CLS–TRAJ variant-level equivalence (5 seeds, 523 variants, 114 clusters). The smallest margin achieving equivalence at $\alpha = 0.05$ is ± 1.52 pp.

Margin ($\pm$ pp)	p-value	Equivalent (α=0.05)?
1.0	$\gg$0.05	No
1.5	0.054	No (borderline)
1.52	**$\sim$0.05**	**Smallest margin**
1.71	0.027	Yes
2.0	0.007	Yes
3.0	4.6×10^{-5}	Yes
5.0	$\ll$0.001	Yes

The minimum equivalence margin is ± 1.52 pp, 0.48 pp below our pre-specified ± 2 pp threshold. The ± 2 pp threshold was set based on deployment relevance rather than observed data: at function-level accuracy (85–88%), a 2 pp per-test difference corresponds to $\sim$38 additional errors across 1,890 tests ($\sim$2–3 problems out of 114), comparable in magnitude to between-seed standard deviation (0.37–0.97 pp) and checkpoint selection noise ($<$1 pp). Differences within this range do not reliably distinguish formulations from training noise in pipeline selection decisions.

J Variant-Level Effect Sizes

Table 14 reports variant-level bootstrap confidence intervals for the repo-level formulation comparisons (n=48 variants). Following Hoenig and Heisey (Hoenig and Heisey, 2001), we report minimum detectable effect (MDE) rather than post-hoc observed power.

Table 14: Variant-level bootstrap analysis (n=48). CLS–CWM uses paired bootstrap (same variants, same seed); CLS–TRAJ uses unpaired bootstrap (per-variant TRAJ predictions not archived for pairing). MDE at n=48: $\sim$25 pp.

Comparison	Config	Δ (pp)	95% CI	p
CLS *vs.* CWM *(paired)*				
	Qwen3-8B	+29.17	[10.42, 47.92]	0.003
	DeepSeek-6.7B	+35.42	[18.75, 50.05]	$<$0.001
CLS *vs.* TRAJ *(unpaired)*				
	Qwen3-4B	$-$4.17	wide	0.740
	Qwen3-8B	+8.33	wide	0.385
	DeepSeek-6.7B	0.00	wide	1.000

At n=48, the MDE is $\sim$25 pp. The CLS–CWM gap (29–35 pp) exceeds this threshold, confirming that n=48 is sufficient to detect the observed divergence. The CLS–TRAJ gaps ($<$10 pp) fall well below the MDE: any true difference is too small to detect at this sample size, consistent with the function-level TOST equivalence (§3.4).

The CLS–TRAJ unpaired bootstrap produces wider confidence intervals than paired analysis would; TRAJ per-variant predictions were not archived in a format that permits pairing. The CLS–CWM paired comparison uses CLS s123 vs. CWM s42 (different seeds), but the CI lower bound of 10.42 pp far exceeds the seed noise observed in function-level experiments ($<$2 pp).

K Multiple Comparison Correction

Table 15 applies Holm-Bonferroni correction (Holm, 1979) to the four primary statistical comparisons across this study (k=4, family-wise α=0.05).

Table 15: Holm-Bonferroni correction for primary comparisons (k=4). Comparisons ranked by raw p-value; each tested against $\alpha/(k - \text{rank} + 1)$.

Rank	Comparison	Raw p	Factor	Adj p	Sig.?
1	Repo: CLS vs CWM	0.003	$\times$4	0.012	✓
2	Func: CLS vs TRAJ (TOST $\pm$2 pp)	0.0071	$\times$3	0.021	✓
3	Repo: CWM vs TRAJ	0.039	$\times$2	0.078	—
4	Repo: CLS vs TRAJ	0.385	$\times$1	0.385	—

The two core claims survive correction: the repo-level CLS–CWM gap (adjusted p=0.012) and function-level CLS–TRAJ equivalence (adjusted p=0.021). The repo-level CWM–TRAJ comparison does not survive correction (adjusted p=0.078); we interpret this difference as directionally consistent but not confirmatory.

L Mixed-Effects Regression

We fit mixed-effects linear probability models (LPM) to test the formulation$\times$complexity interaction formally. We choose LPM because coefficients are directly interpretable as probability differences (percentage points), consistent with the reporting units in the main text; at these sample sizes ($n\approx$56K cross-complexity, repo-level pooled cross-model) and base rate range (0.6–0.9), LPM and logistic regression yield identical significance conclusions (Angrist and Pischke, 2009).

Cross-complexity model (DeepSeek-Coder). `correct ~ formulation × complexity + (1|problem_id)`, n=56,480 observations (CLS + CWM $\times$ function + repo), 139 problem clusters. The CWM main effect at function level is non-significant ($\hat{\beta}$= 0.002, z= $-$ 0.25, p=0.805); the

CWM×repo interaction is highly significant ($\hat{\beta}= -0.219$, $z= -23.7$, $p<0.001$). CWM and CLS are statistically indistinguishable at function level ($\Delta= -0.2$ pp, $p=0.805$) but diverge by 22 pp at repo level, confirming the complexity-dependent divergence is not an artifact of aggregation.

Repo-level cross-model model. `correct ~ formulation × model + (1|problem_id)`, both model families, repo level only. CWM deficit: $\hat{\beta}= -0.22$ (-22 pp, $z= -64.4$, $p<0.001$). Model effect: +1.1 pp for Qwen3 ($z=3.1$, $p=0.002$). Formulation×model interaction: +1.6 pp less CWM degradation on Qwen3 ($z=2.8$, $p=0.006$). Formulation explains >95% of the predictable variance; model identity modulates gap magnitude without changing direction. The 25-cluster random intercept structure limits effective degrees of freedom for the random effects; inferences on fixed effects (formulation, complexity) are robust as they pool across the full observation count.

M CWM Parseable Bias Analysis

The default-to-pass assignment for unparseable CWM outputs is not random with respect to instance difficulty. Unparseable instances have significantly higher pass rates than parseable ones (74.6% vs. 68.3%, χ^2 $p = 5.9 \times 10^{-10}$), and CLS models achieve higher accuracy on unparseable instances (86.4% vs. 79.1%, $p = 3.3 \times 10^{-18}$). This indicates that CWM's parsing failures concentrate on simpler, pass-heavy instances where the majority-class default happens to be correct more often. Cross-model Jaccard overlap of unparseable instance sets is 0.46, confirming this pattern is instance-driven (specific inputs consistently cause parsing failure across models) rather than stochastic. The net effect is that CWM's reported accuracy is inflated by default-correct assignments on easy instances, while its effective discriminative accuracy on harder instances is substantially lower than the aggregate figure suggests.

N Threats to Validity (Full Analysis)

We identify five potential confounds and assess their impact on our conclusions.

Checkpoint selection. Val-based selection is unavailable for runs with NaN validation loss (all CWM repo-level runs, DeepSeek CLS s123), which default to the final epoch checkpoint. A checkpoint ablation on Qwen3-8B CLS s42 quantifies this confound: epoch 1 (ckpt-2200) achieves 85.34%, val-selected (ckpt-3600) 84.48%, and final epoch (ckpt-6633) 85.29%—a <1 pp range. The val-selected checkpoint is the lowest, indicating no upward bias from checkpoint selection. The Qwen3-8B TRAJ comparison (epoch 2: 72.92% vs. epoch 3: 70.83%) shows a slightly larger ∼2 pp shift.

Evaluation sequence length. CWM accuracy varies by ∼9 pp across evaluation sequence lengths for the same checkpoint (seq2048: ∼65% vs. seq4096: ∼56%), driven by unparseable-rate differences that propagate through majority-class default assignment. CLS is invariant to this parameter. Cross-model CWM comparisons use different eval lengths (Qwen3: 8192, DeepSeek: 4096) and are not directly comparable in absolute terms. A matched-sequence evaluation shows ∼1.5 pp difference for CLS (Appendix B), insufficient to explain the 19–26 pp gap. Furthermore, increasing Qwen3-8B CWM evaluation from seq4096 to seq8192 *raises* the unparseable rate (63.7%→73.8%) while changing accuracy by only 0.81 pp (64.36%→65.17%); giving the model more generation budget worsens format compliance rather than improving it, confirming that format collapse is a structural bottleneck rather than a truncation artifact. Practitioners must standardize evaluation sequence length for generation-based formulations.

Preprocessing. DeepSeek CLS accuracy ranges from 82.61% (s42, drop preprocessing) to 74.93% (s123, truncate preprocessing), a 7.68 pp spread that conflates preprocessing strategy, checkpoint selection method, and seed. The two strategies produce different effective training sets (67,343 vs. 70,748 samples) and different checkpoint selection regimes. The qualitative ranking CLS > CWM holds under both strategies (Table 4).

TRAJ training regime. TRAJ uses a different training configuration: smaller training set (∼400 repo-level variants vs. ∼70K per-test instances), higher learning rate (2×10^{-4} vs. 5×10^{-5}), and variant-level labels. This asymmetry is inherent to the formulation (trajectory-level labels are only available at variant granularity), but precludes training-data-controlled comparison at repo level. The downsample ablation confirms the CLS for-

mulation advantage persists at matched data volume (80.74% vs. 72.92%; Appendix B), though it controls for training volume but not label granularity (per-test vs. variant-level). Crucially, TRAJ achieves statistical equivalence with CLS at function level ($\leq$1.3 pp, §3.2), demonstrating the formulation itself is viable; the repo-level gap is more consistent with data scarcity ($\sim$400 variants vs. $\sim$70K per-test instances) than with a formulation-level limitation.

CWM scale and pretraining. Our CWM baseline uses 4–8B LoRA fine-tuning without execution trace pretraining. The original Code World Model (FAIR CodeGen team et al., 2025) uses 32B parameters with mid-training on execution traces, a fundamentally different training regime. At larger scale or with execution-trace pretraining, the generation bottleneck observed here may diminish: larger models produce more coherent long-form outputs, and execution trace exposure may improve output format compliance. This is the primary limitation: all repo-level CWM conclusions are conditional on the $\leq$8B LoRA regime.

Despite these confounds, the repo-level CLS–CWM gap (19–26 pp) exceeds any single assessed confound. Even under worst-case compounding ($\sim$11.5 pp), the gap retains $\geq$7.5 pp margin; confounds predominantly disadvantage CWM. However, correlated confounds and limited seed coverage mean definitive causal attribution requires multi-seed replication. At n=48, the minimum detectable effect (MDE) is $\sim$25 pp (Hoenig and Heisey, 2001); the CLS–CWM gap (29–35 pp) exceeds this threshold, while the CLS–TRAJ gap ($<$10 pp) falls below it, consistent with our TOST equivalence finding. Holm-Bonferroni correction across the four primary comparisons (k=4) confirms both the CLS–CWM gap (adjusted p=0.012) and function-level equivalence (adjusted p=0.021) survive; the repo-level CWM–TRAJ comparison is directionally consistent but does not survive correction (adjusted p=0.078; Table 15). The consistent direction across both model families and all tested configurations supports a formulation-level explanation, though definitive causal attribution requires future experiments addressing each confound individually.

O Extended Limitations

The same-size 4B comparison uses a single seed; CLS 4B seed 123 (val-best ckpt-800) achieves only 75.33% test accuracy (+2.4 pp above majority baseline); val-best selection at an early checkpoint (step 800) amplifies the val-test distribution shift observed across all configurations. The CWM 8B comparison is also single-seed. The validation and test sets have zero problem overlap but differ in mutation type composition (Table 12): `operator_swap` shifts from 19.1% (val) to 34.3% (test), the largest single-type distribution mismatch (+15.2 pp). All repo-level results use val-selected checkpoints, and checkpoint selection variance could contribute to observed differences. Our training data is synthetically constructed from mutation operators rather than real developer patches. While mutation testing literature establishes that mutants are reasonable proxies for real faults in test effectiveness studies (Andrews et al., 2005; Just et al., 2014), these findings concern mutation *adequacy* (whether tests that kill mutants also detect real faults), not whether verifiers trained on mutants transfer to real patches. However, the coupling effect underlying these results supports *relative ranking* validity: a domain shift from synthetic mutations to real bugs may affect absolute accuracy levels, but our core finding—that formulation choice matters at repo level but not function level—concerns relative performance ordering, which is expected to be preserved under distributional shifts that affect all formulations equally. The synthetic-to-real transfer gap for absolute accuracy remains untested. The TRAJ training configuration differs from CLS/CWM: smaller training set ($\sim$400 vs. $\sim$70K instances), different learning rate (2×10^{-4} vs. 5×10^{-5}), and variant-level evaluation (n=48 vs. per-test n=9,784). The function-level TOST equivalence ($p = 0.007$, $\pm$2 pp) is computed on 523 matched variant-level instances across 5 seeds; 2 of 5 seeds use NF4 quantization. Restricting to three bfloat16 seeds yields a comparable mean gap ($-$0.45 pp) but wider confidence interval (df=2); the TOST p-value increases from 0.007 to 0.128, which reflects reduced statistical power rather than a shift in the point estimate. The NF4 seeds contribute precision (narrowing the CI) without shifting the mean, consistent with precision being comparable across both quantization levels at function level. Repo-level CLS–TRAJ

comparison should be interpreted with caution due to configuration and granularity differences.